WORKSHEET

Chapter 10: Sampling distributions

You need access to a random sample generator for this worksheet, such as a CAS calculator, suitable spreadsheet or computer program.

1 CAS Identical cards are numbered 1 through 9 and paced in a hat. A card is randomly drawn and its number noted. The card is then replaced.

a Use the full sample space list {1, 2, 3, 4, 5, 6, 7, 8, 9} to simulate 150 samples of 50 for the number shown on the card drawn and find the means of each sample.

b Draw a graph of the results.

c Comment on the results.

2 Two die are rolled and the absolute value of the difference of the numbers is calculated.

a Show the sample space as a 6×6 table.

b Use the full sample space list to simulate a sample of 36 results and draw the graph.

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c Use the full sample space list to simulate 120 samples of 30 results and find the means of each sample.

d Draw a graph of the results.

e Compare to the graphs.

3 CAS **a** Simulate a sample of 40 items from a uniform distribution on the interval [0, 10] and draw a graph.

b Simulate 100 samples of 40 items from a uniform distribution on the interval [0, 10] and find the means of the samples.

c Draw a graph of the sampling distribution.

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d Comment on the graphs.

4 CAS **a** Simulate a sample of 80 items from a uniform distribution on the interval [20, 40] and draw a graph.

b Simulate 200 samples of 80 items from a uniform distribution on the interval [20, 40] and find the means of the samples.

c Draw a graph of the sampling distribution.

d Comment on the graphs.

5 CAS **a** Simulate a sample of 25 items from a uniform distribution on the interval [15, 30] and draw a graph.

b Simulate 100 samples of 25 items from a uniform distribution on the interval [15, 30] and find the means of the samples.

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c Draw a graph of the sampling distribution.

d Simulate 100 samples of 80 items from a uniform distribution on the interval [15, 30] and find the means of the samples.

e Draw a graph of the sampling distribution.

f Comment on the graphs.

6 CAS **a** Simulate a sample of 20 items from a binomial distribution with $p = 0.4$ and $n = 15$ and draw a graph.

b Simulate 100 samples of 20 items from a binomial distribution with $p = 0.4$ and $n = 15$ and find the means.

c Draw a graph of the sampling distribution.

d Simulate 100 samples of 80 items from a binomial distribution with $p = 0.4$ and $n = 15$ and find the means.

e Draw a graph of the sampling distribution.

f Comment on the graphs.

7 CAS **a** Simulate a sample of 15 items from a binomial distribution with $p = 0.8$ and $n = 7$ and draw a graph.

b Simulate 200 samples of 15 items from a binomial distribution with $p = 0.8$ and $n = 7$ and find the means.

c Draw a graph of the sampling distribution.

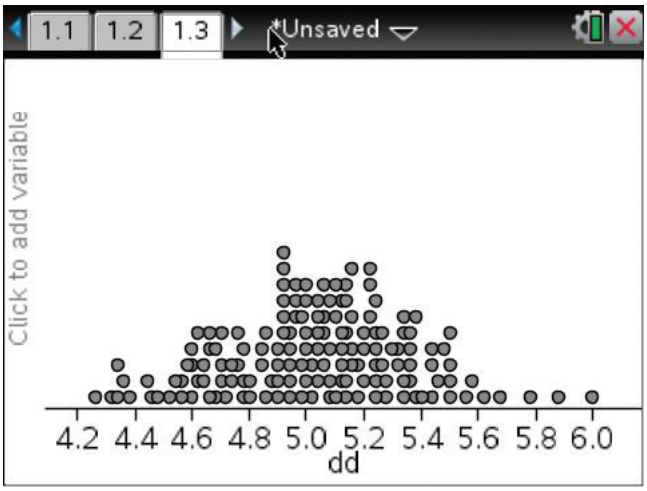
d Simulate 200 samples of 150 items from a binomial distribution with $p = 0.8$ and $n = 7$ and find the means.

e Draw a graph of the sampling distribution.

f Comment on the graphs.

Answers

1 **a, b** Similar to:

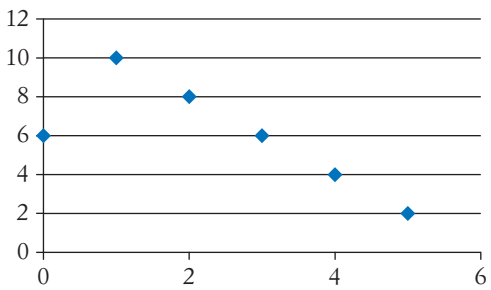


c The graph is not a uniform distribution like the original. It is more bell-shaped and extends only from about 4.2 to 6.0.

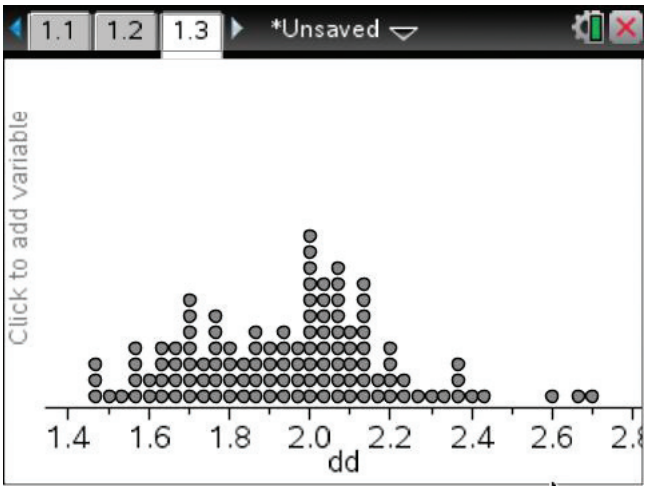
2 **a**

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

b

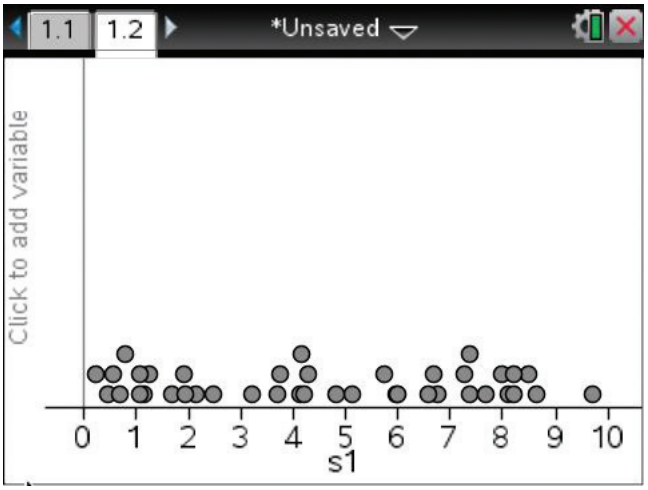


c, d Similar to:

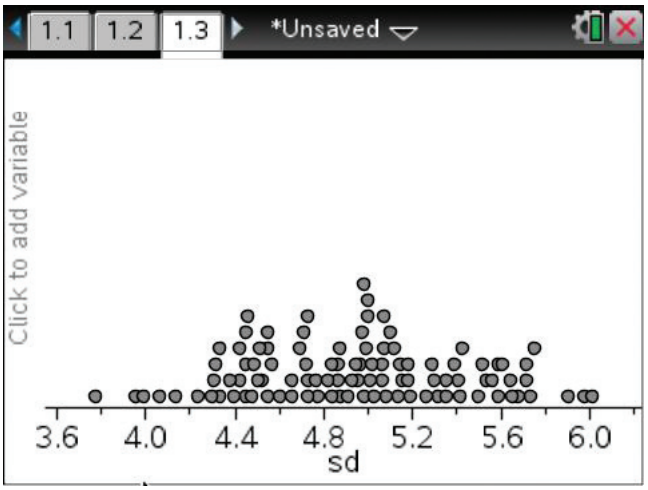


e The sampling distribution is spread from about 1.4 to 2.7 with a peak at about 2.0. It is bell-shaped.

3 a Similar to:

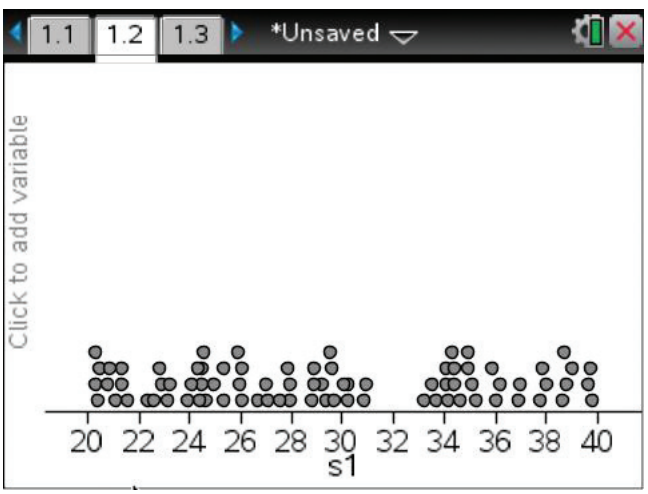


b, c Similar to:

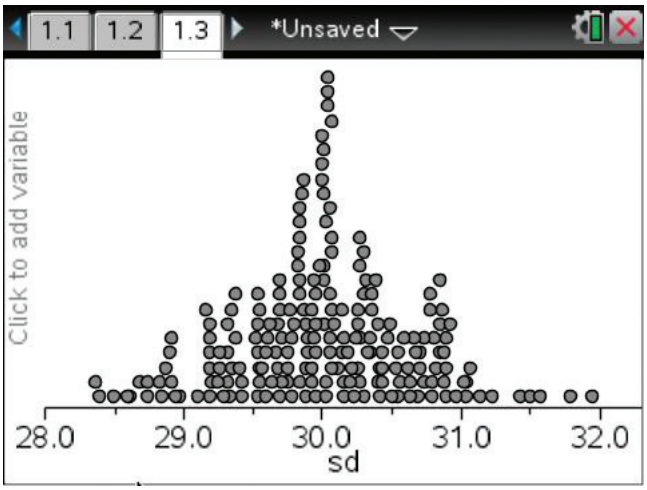


d The single sample is distributed from 0 to 10, but the sampling distribution is only from about 3.8 to 6 with a peak about 5 in a bell-shaped curve.

4 a Similar to:

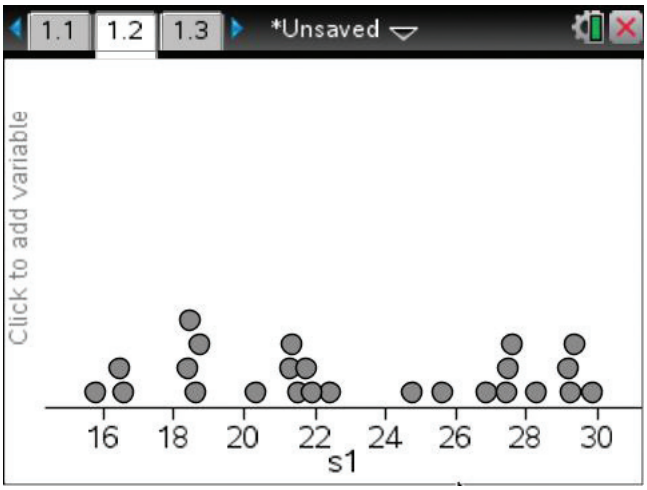


b, c Similar to:

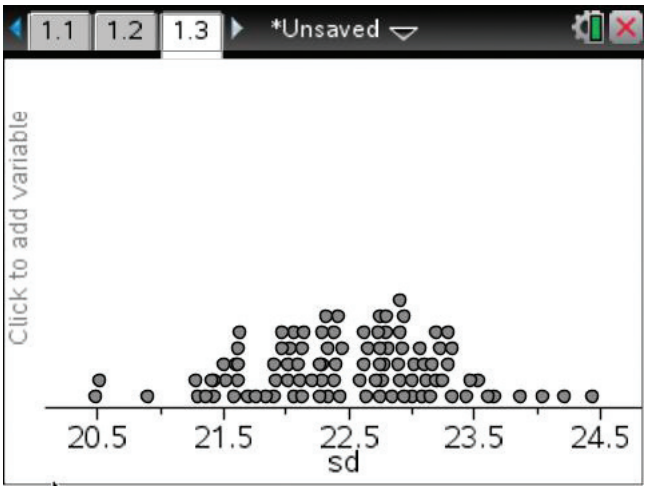


d The single sample is distributed from 20 to 40, but the sampling distribution is only from about 28.3 to 32 with a peak at about 30 in a bell-shaped curve.

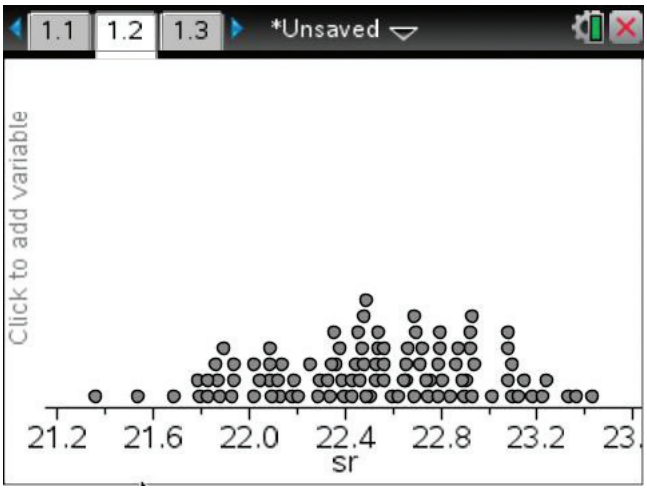
5 a Similar to:



b, c Similar to:

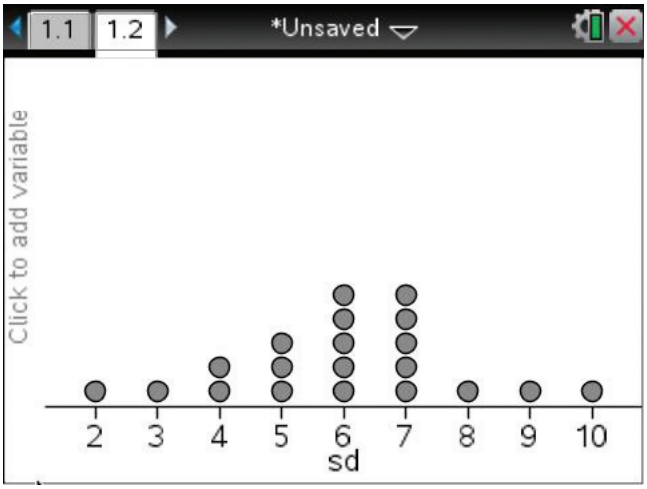


d, e Similar to:

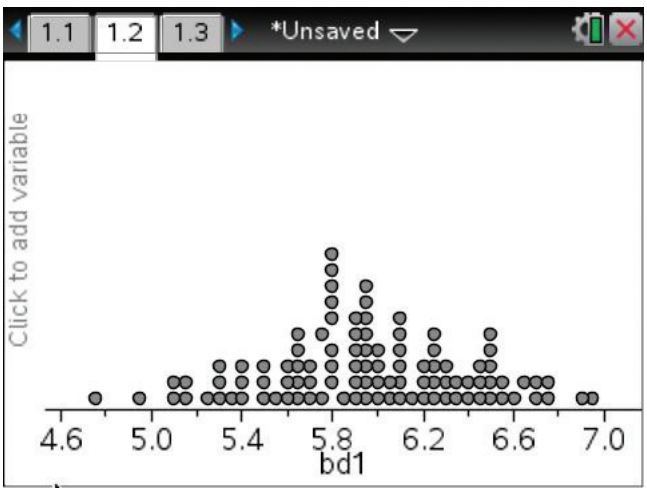


f The single sample is distributed from about 15 to 30, but the sampling distributions are only from about 20.5 to 24.5 and 21.4 to 23.6 with peaks about 23 and 22.6 respectively in bell-shaped curves. The larger sample size has a more concentrated distribution.

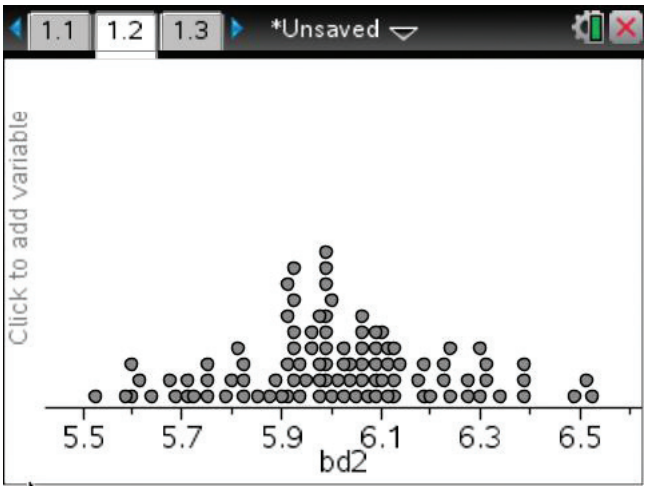
6 a Similar to:



b, c Similar to:

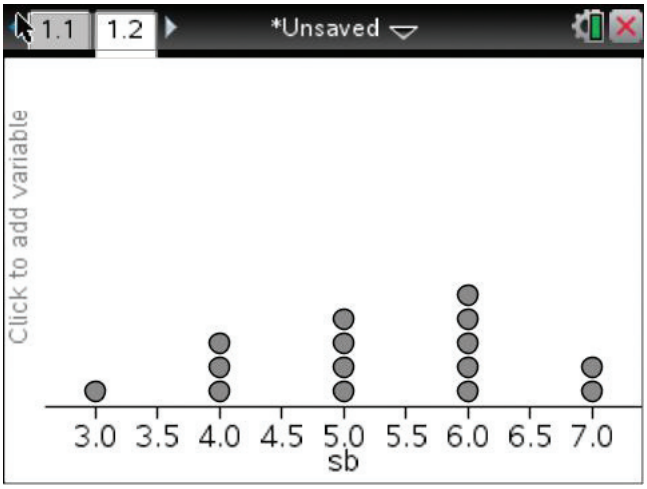


d, e Similar to:

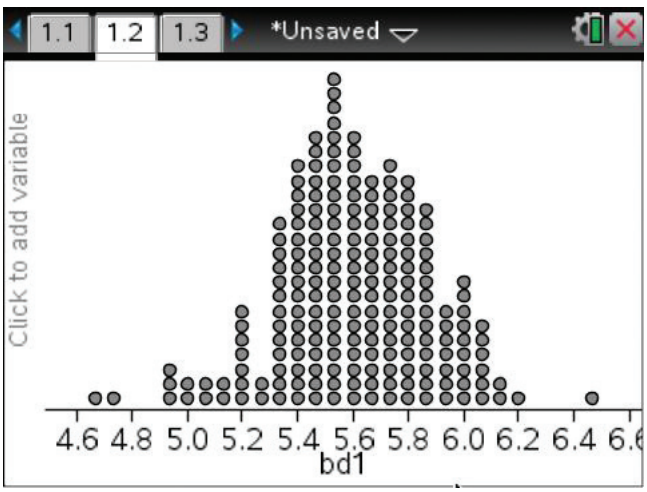


f The single sample is unevenly distributed from about 2 to 10, but the sampling distributions are only from 4.8 to 7 and 5.5 to 6.5 with peaks about 6 in bell-shaped curves. The larger sample size has a more concentrated distribution.

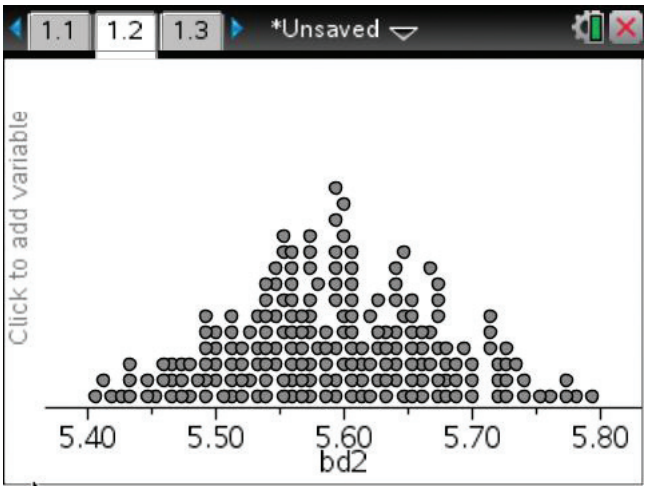
7 a Similar to:



b, c Similar to:



d, e Similar to:



f The single sample is unevenly distributed from about 3 to 7, but the sampling distributions are only from 4.6 to 6.5 and 5.4 to 5.8 with peaks about 5.6 in bell-shaped curves. The larger sample size has a more concentrated distribution.

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